

Straightening Laws in Algebraic Complexity

Anakin Dey

What is Algebraic Complexity?

Question

How hard is it to compute polynomials?

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Can we find *explicit* polynomials which are hard to compute?

Algebraic Circuits

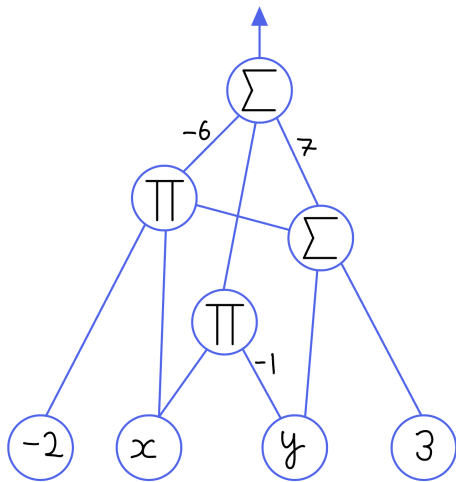
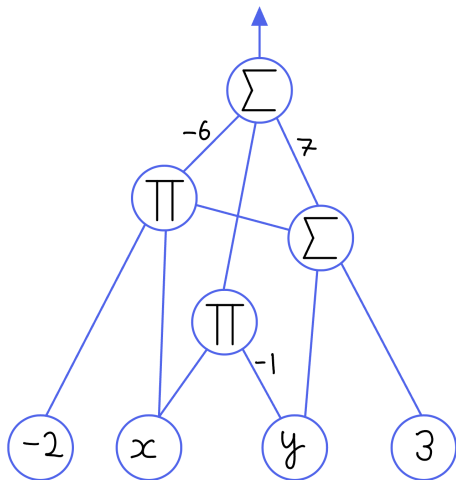


Figure: $-6(-2x \cdot (y + 3)) - xy + 7(y + 3)$

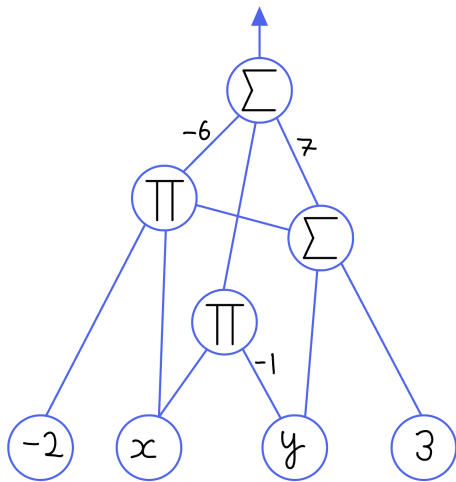
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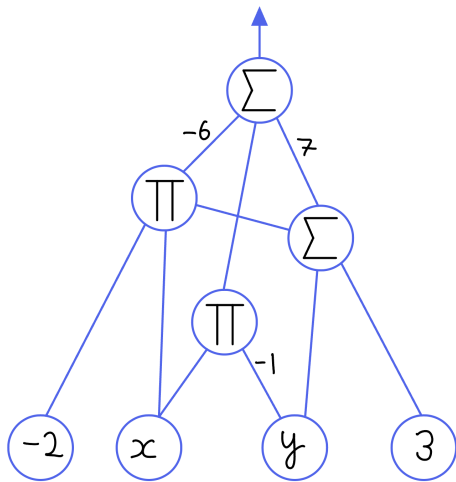


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We have two measures of complexity:

- *Size*: The number of edges
- *Depth*: The length of the longest input-output path

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Horner's Rule

Let $f(x) = \sum_{i=0}^d a_i x^i \in \mathbb{F}[x]$ be a univariate polynomial.

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- The number of additions is optimal [Ost54].
- The number of multiplications is optimal [Pan66].

Hard Polynomials Exist

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Let $\mathbb{F}$ be a field of characteristic zero. Then “most” n -variate polynomials of degree $\leq d$ require circuits of size $N := \binom{n+d}{d}$.

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- $\dim \operatorname{im} \Phi \leq s$, and there are only finitely many unlabeled circuits.
- Thus, the dimension over all possible unlabeled circuits is at most s .
- For the image to be Zariski-dense, we must have that $s \geq N$.

Determinant versus Permanent

Conjecture

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Suppose that A is an $m \times m$ matrix with entries in $\mathbb{F}[X]_{\leq 1}$ such that

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Then m must be *superpolynomial* in n .

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We say a circuit *border computes* $g(\overline{x})$ over $\mathbb{F}(\varepsilon)$ if it computes

$$g(\overline{x}) + \varepsilon^q \tilde{g}(\overline{x}, \varepsilon) \in \mathbb{F}[[\varepsilon]][\overline{x}], \quad q \geq 1.$$

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We **cannot** set $\varepsilon = 0$ since we may divide by ε .

What does Border Computation Get Us?

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What is the power of border computation?

Is it strictly necessary? Can we prove something in the border setting and *deborder* the result to get an exact computation?

Border Waring Rank

Definition

The *Waring Rank*, $\text{WR}(f)$, of a degree d form $f(\bar{x})$ over $\mathbb{C}$ is the minimum r such that there exist linear forms $\ell_1(\bar{x}), \dots, \ell_r(\bar{x})$ such that

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$\implies \{ f(x, y) \in \mathbb{C}[x, y]_{\leq d} \mid \text{WR}(f) = 2 \}$ is *not* Zariski-closed.

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Suppose $f \in \langle g_1, \dots, g_k \rangle$. How does the complexity of f compare to the complexity of the generators $g_1, \dots, g_k$?

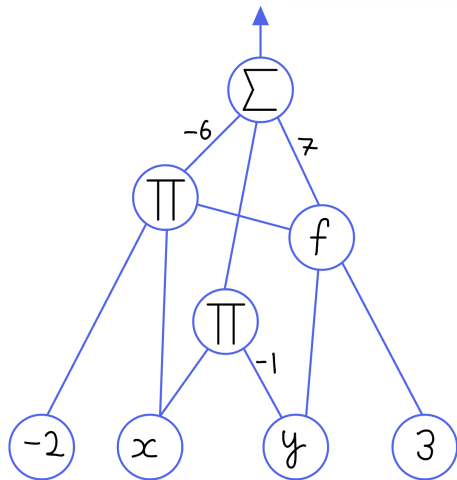
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We can allow a fixed polynomial $f(\bar{x})$ to be computed with unit cost by allowing it to function as its own gate.

Figure: $-6(-2x \cdot f(y, 3)) - xy + 7f(y, 3)$

Principal Ideals

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Suppose $0 \neq f \in \langle g \rangle$. Does g have a small f -oracle circuit?

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This is open even if we ask that $\{g_1, g_2\}$ are a Gröbner basis or form a regular sequence.

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- However, $\det_n = \text{perm}_n + (\det_n - \text{perm}_n) \in I$ is known to have a circuit of size $\text{poly}(n)$ computing it.
- Thus, our question is asking us to resolve $\det_n$ versus perm_n .

Adding $\det_n$ to the set of generators remedies this issue.

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Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I_{n,m,r}^{\det}$ be the *determinantal ideal* generated by the $r \times r$ minors of X .

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- However, these circuits are *not* constant-depth.
- The oracle allows *constant depth* computation of the determinant.

Closure Results in Determinantal Ideals

Theorem ([AF22, Theorem 1.1])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $O(n^2 m^2)$ that *border computes* the $t \times t$ determinant for some $t = O(r^{1/3})$.

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Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = O(r^{1/3})$.

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Main Tools:

- We use *Straightening Laws* from Invariant Theory to express $f(X)$ in a standard basis indexed by Young tableau, and leverage the combinatorics to talk about specific terms.
- To get a circuit for a specific basis term, we use *Interpolation* as well as a novel application of the *Isolation Lemma*.

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- Working with the monomial basis is hard since determinants have exponentially many terms.
- Thus, we will work with a different basis for $I_{n,m,r}^{\det}$ which is described combinatorially with Young tableau.

A Natural Generating Set for Determinantal Ideals

Definition (Bideterminants)

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$$(S \mid T) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 4 & 5 \\ \hline 3 & 6 & & \\ \hline 4 & & & \\ \hline \end{array} \mid \begin{array}{|c|c|c|c|} \hline 1 & 3 & 5 & 6 \\ \hline 2 & 7 & & \\ \hline 3 & & & \\ \hline \end{array} \right).$$

A Natural Generating Set for Determinantal Ideals

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Then the *bideterminant* $(S \mid T)(X)$ is given by

$$(S \mid T)(X) = \begin{pmatrix} x_{1,1} & x_{1,3} & x_{1,5} & x_{1,6} \\ x_{2,1} & x_{2,3} & x_{2,5} & x_{2,6} \\ x_{4,1} & x_{4,3} & x_{4,5} & x_{4,6} \\ x_{5,1} & x_{5,3} & x_{5,5} & x_{5,6} \end{pmatrix} \begin{pmatrix} x_{3,2} & x_{3,7} \\ x_{6,2} & x_{6,7} \end{pmatrix} (x_{4,3}).$$

A Natural Generating Set for Determinantal Ideals

Lemma

A degree d monomial $\prod_{i=1}^d x_{r_i, c_i}$ is given by a bideterminant:

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Thus, the bideterminants span $\mathbb{F}[X]$.

A Natural Generating Set for Determinantal Ideals

Theorem ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3])

Let $(S \mid T)(X)$ be a bideterminant of shape σ .

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Let $(S \mid T)(X)$ be a bideterminant of shape σ . Then $(S \mid T)(X)$ can be uniquely expressed as a linear combination

$$(S \mid T)(X) = \sum_{(A|B)} c_{A,B} (A \mid B)(X), \quad c_{A,B} \in \mathbb{Z}$$

where A and B are *conjugate semistandard*: strictly increasing along rows, weakly increasing down columns.

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Furthermore, if $(A \mid B)$ has shape τ , then $\tau \geq \sigma$.

A Natural Generating Set for Determinantal Ideals

Since $I_{n,m,r}^{\det}$ is generated by the $r \times r$ minors, the bideterminants in our basis should have at least one determinant of length at least r .

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Since $I_{n,m,r}^{\det}$ is generated by the $r \times r$ minors, the bideterminants in our basis should have at least one determinant of length at least r .

Proposition ([BC03, Corollary 1.7])

The ideal $I_{n,m,r}^{\det}$ has as basis the conjugate semistandard bideterminants of shape σ whose first row has length at least r .

Shuffle Products

$$\left(\begin{array}{|c|c|c|c|c|c|} \hline i_1 & \cdots & i_p & i_{p+1} & \cdots & i_s \\ \hline l_1 & \cdots & l_q & l_{q+1} & \cdots & l_t \\ \hline \end{array} \quad \bigg| \quad \begin{array}{|c|c|c|} \hline j_1 & \cdots & j_s \\ \hline m_1 & \cdots & m_t \\ \hline \end{array} \right)$$

$$= \sum_{\sigma} \operatorname{sgn}(\sigma) \left(\begin{array}{|c|c|c|c|c|c|} \hline \sigma(i_1) & \cdots & \sigma(i_p) & i_{p+1} & \cdots & i_s \\ \hline \sigma(l_1) & \cdots & \sigma(l_q) & l_{q+1} & \cdots & l_t \\ \hline \end{array} \quad \bigg| \quad \begin{array}{|c|c|c|} \hline j_1 & \cdots & j_s \\ \hline m_1 & \cdots & m_t \\ \hline \end{array} \right)$$

Where the summation is over all permutations of $\{i_1, \dots, i_p, l_1, \dots, l_q\}$ such that $\sigma(i_1) < \dots < \sigma(i_p)$ and $\sigma(l_1) < \dots < \sigma(l_q)$.

Laplace Expansion

$$\left(\begin{array}{|c|c|c|c|c|c|} \hline i_1 & \cdots & i_p & i_{p+1} & \cdots & i_s \\ \hline l_1 & \cdots & l_q & l_{q+1} & \cdots & l_t \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline j_1 & \cdots & j_s \\ \hline m_1 & \cdots & m_t \\ \hline \end{array} \right)$$

$$= \sum_{\tau, \mu} \text{sgn}(\tau) \text{sgn}(\mu) \left(\begin{array}{|c|c|c|c|c|c|} \hline i_1 & \cdots & i_p & l_1 & \cdots & l_q \\ \hline i_{p+1} & \cdots & i_s & & & \\ \hline l_{q+1} & \cdots & l_t & & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|c|c|c|} \hline \tau(j_1) & \cdots & \tau(j_p) & \mu(m_1) & \cdots & \mu(m_t) \\ \hline \tau(j_{p+1}) & \cdots & \tau(j_s) & & & \\ \hline \mu(m_{q+1}) & \cdots & \mu(m_t) & & & \\ \hline \end{array} \right)$$

Where the summation is over all permutations of $\{j_1, \dots, j_p, j_{p+1}, \dots, j_s\}$ such that $\tau(j_1) < \dots < \tau(j_p)$ and $\tau(j_{p+1}) < \dots < \tau(j_s)$ and similar for μ .

Example Straightening

Consider the following nonstandard tableau

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right)$$

Example Straightening

Shuffle:

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) + \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right)$$

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Laplace:

$$\begin{aligned} \left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) &= \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 1 & 2 \\ \hline 3 & & \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 3 & 1 & 2 \\ \hline 1 & & \\ \hline \end{array} \right) \\ &= - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 1 & & \\ \hline \end{array} \right) \end{aligned}$$

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We notice that our unstraightened rows appear in the first expression so that we can solve for it.

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Thus,

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline 2 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 1 & & \\ \hline 1 & & \\ \hline \end{array} \right)$$

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All terms that appear have the same content and they are either the same shape with a lexicographically smaller filling or they have a longer shape.

Main Theorem

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. Then there is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant for some $t = \text{poly}(n)$.

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Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = O(r^{1/3})$.

Main Theorem

Theorem

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then, there exists a depth-three f -oracle circuit of size $O(n^2 m^2 d^3 (n^3 + m^3))$ computing $(K_\sigma \mid K_\sigma)(X)$, where $\sigma_1 \geq r$ and $|\sigma| \leq d$.

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Definition (Canonical Bideterminant [dCEP80])

Let $\sigma = (4, 2, 1)$:

$$(K_\sigma \mid K_\sigma)(X) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & & \\ \hline 1 & & & \\ \hline \end{array} \mid \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & & \\ \hline 1 & & & \\ \hline \end{array} \right) (X)$$

= product of upper-left minors.

Proof Idea

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$$f(X) = \sum_{k \in I} c_k(S_k \mid T_k)(X).$$

Goal: Use $f(X)$ as an oracle to compute a canonical bideterminant.

Linear Operators

Definition (Substitution operator [dCEP80])

For $i < j$, the operator $\text{Sub}_{i \rightarrow j}$ takes a tableau S and returns the tableau S' formed by taking each row of S which has an i but not a j , replacing that i with j , and sorting the row in increasing order.

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$$\text{Sub}_{2 \rightarrow 4} \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & & \\ \hline 1 & & & \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 4 & & \\ \hline 1 & & & \\ \hline \end{array}$$

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Let $h_i^j(S)$ be the number of times i is replaced by j in S .

Linear Operators

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Let S be a conjugate semistandard tableau of shape σ such that S has entries of value at most n .

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Let $\sigma = (4, 2, 1)$ and $n = m = 6$:

$$\begin{aligned} (\overline{K}_\sigma \mid \overline{K}_\sigma)(X) &= \left(\begin{array}{|c|c|c|c|} \hline 3 & 4 & 5 & 6 \\ \hline 5 & 6 & & \\ \hline 6 & & & \\ \hline \end{array} \mid \begin{array}{|c|c|c|c|} \hline 3 & 4 & 5 & 6 \\ \hline 5 & 6 & & \\ \hline 6 & & & \\ \hline \end{array} \right) (X) \\ &= \text{product of lower-right minors.} \end{aligned}$$

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$$(S \mid T)(E_{i,j}(\lambda)X)$$

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$$(S \mid T)(E_{i,j}(\lambda)X) = \pm \lambda^{h_i^j(S)} (\text{Sub}_{i \rightarrow j}(S) \mid T)(X)$$

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Let λ be a new indeterminate and let $(S \mid T)$ be a bitableau.

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm \lambda^{h_i^j(S)} (\text{Sub}_{i \rightarrow j}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S' \in \mathcal{C}_{i \rightarrow j}^h(S)} \pm (S' \mid T)(X).$$

For $0 \leq h \leq h_i^j(S) - 1$, let $\mathcal{C}_{i \rightarrow j}^h(S)$ be the set of tableaux of shape σ obtained by changing i to j at exactly h rows of S which contain i but not j and reordering those rows to be increasing.

Applying all Substitution Operators

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Lemma

$$(\bar{K}_\sigma \mid \bar{K}_\sigma)(J_n X J_m) = \pm (K_\sigma \mid K_\sigma)(X).$$

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Lemma ([DG25, Corollary 3.10])

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Distinguishing Terms

$$D = \text{diag}(y_1 v, y_2 v^2, \dots, y_n v^n) \in \mathbb{F}[Y, v]^{n \times n}$$

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Lemma

Let $(S \mid T)(X)$ be a bideterminant. Then

$$(S \mid T)(DXD') = Y^{\bar{s}} Z^{\bar{t}} v^{|S|+|T|} \cdot (S \mid T)(X).$$

- $(\bar{s}, \bar{t})$ is the multidegree of $(S \mid T)(X)$: $\text{mdeg}(x_{i,j}) = e_i \oplus e_j$
- $|S|$ is the sum of the entries in S .

Isolating Terms

Lemma

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Let $g = f(MJ_nDXD'J_mN) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, v]$.

$$\text{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X).$$

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We now *interpolate* to get an f -oracle circuit computing $\text{coeff}_{v^{d_{\min}}}(g)$.

Interpolation

Definition

Consider a degree d polynomial $g(\bar{x}, t) = \sum_{i=0}^d \text{coeff}_{t^i}(g)t^i$.

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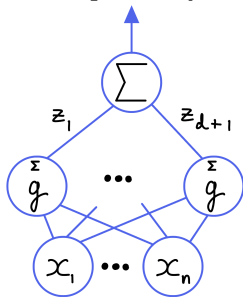
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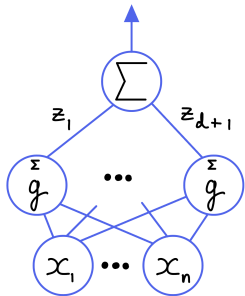
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Proof

Express *Lagrange interpolation* as a circuit.

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$$g(X) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) \in \mathbb{F}[\Lambda, \Xi, Y, Z][X].$$

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Problem: Interpolating with respect to multiple variables is *expensive*.

Issues with Interpolation

Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

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If a circuit border computes $g(\bar{x})$, then it computes

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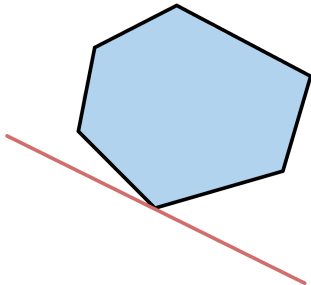
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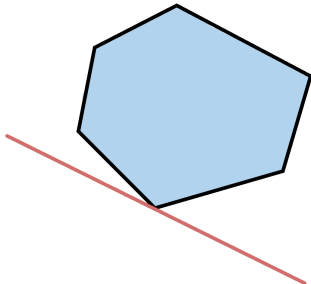


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Moreover, if the linear equations have bounded integer coefficients, then evaluation at *small, random* values has a unique minimum.

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Then, with probability at least $1 - \varepsilon$, there is a unique form in $\mathcal{L}$ of minimum value at $z_1, \dots, z_\ell$.

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $g(x_1, \dots, x_\ell)$ such that the individual degree of each x_i in g is at most K :

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Pfaffian Analogues

Definition

If X is a $2n \times 2n$ skew-symmetric matrix,

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Theorem ([DG25, Theorem 1.6])

Let $f \in I_{2n, 2r}^{\text{pfaff}}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, \deg(f))$ that *exactly computes* the $t \times t$ Pfaffian for some $t = O(r^{1/3})$.

Symmetric Analogue?

Let $x_{i,j}$ be indeterminates for $1 \leq i \leq j \leq n$ and let $x_{j,i} := x_{i,j}$ for $1 \leq j < i \leq n$. Thus, $X = (x_{i,j})_{1 \leq i,j \leq n}$ is a symmetric $n \times n$ matrix.

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Conjecture

Let $I_{n,r}^{\text{sym}} \subseteq \mathbb{F}[X]$ be the ideal generated by the $r \times r$ minors of X . Let $f \in I_{n,r}^{\text{sym}}$ be a nonzero polynomial. Then there exists a constant-depth f -oracle circuit of size $\text{poly}(n, \deg(f))$ that computes the $t \times t$ determinant of a symmetric matrix for $t = \text{poly}(n)$.

Partial Progress

Theorem

Let $f(X) \in I_{n,r}^{\text{sym}}$ be a nonzero polynomial of degree d . Then, there exists a depth-three f -oracle circuit of size $O(d^3 n^7)$ computing $(K_\sigma \mid K_\sigma)(X)$, where $\sigma_1 \geq r$ and $|\sigma| \leq d$.

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Our main roadblock for fully resolving the symmetric case is the reduction from a product of determinants to a single determinant.

Roadblocks for the Permanent

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Is there an analogue to our results for the ideal of permanents of a matrix?

The main roadblock is that we have no analogue of the following result:

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm \lambda^{h_i^j(S)} (\text{Sub}_{i \rightarrow j}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S' \in \mathcal{C}_{i \rightarrow j}^h(S)} \pm (S' \mid T)(X).$$

Algebras with Straightening Laws / Hodge Algebras

Definition ([Eis80, dCEP82])

Let A be a ring, $H \subseteq A$ is a poset.

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Let R be a ring, A an R -algebra, and $H \subseteq A$ a finite poset which generates A as an R -algebra. A is an *Algebra with a Straightening Law* if

- A is a free R -module with a basis given by the standard monomials
- if $\alpha, \beta \in H$ are incomparable, and if

$$\alpha\beta = \sum_{i=1}^m r_i \gamma_{i,1} \cdots \gamma_{i,k_i}$$

where $0 \neq r_i \in R$ and each $\gamma_{i,1} \cdots \gamma_{i,k_i}$ is a standard monomial, we must have that for all $1 \leq i \leq m$ that $\gamma_{i,1} \leq \alpha, \beta$.

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These relations in the second requirement are known as the *straightening relations* or *straightening laws*.

Hankel Ideals

Let $\mathbb{F}$ be a field. Let x_i be indeterminates for $1 \leq i \leq n$. For $1 \leq j \leq n$, let X_j be the following Hankel matrix:

$$\begin{pmatrix} x_1 & x_2 & x_3 & \cdots & x_{n-j+1} \\ x_2 & x_3 & x_4 & \cdots & x_{n-j+2} \\ x_3 & x_4 & x_5 & \cdots & x_{n-j+3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_j & x_{j+1} & x_{j+2} & \cdots & x_n \end{pmatrix}.$$

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For $1 \leq r \leq \min \{ j, n - j + 1 \}$ let $I_r(X_j)$ be the ideal generated by $r \times r$ minors of X_j .

Hankel Ideals

The ideals $I_r(X_j)$ do not have the structure of an algebra with a straightening law [Con98].

In particular, the product of standard monomials need not be standard.

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In particular, the product of standard monomials need not be standard.

We still conjecture the following:

Conjecture

Let $I_r(X_j) \subseteq \mathbb{F}[x_1, \dots, x_n]$ be the ideal generated by the $r \times r$ minors of X_j . Let $f \in I_r(X_j)$ be a nonzero polynomial. Then there exists a constant-depth f -oracle circuit of size $\text{poly}(n, \deg(f))$ that computes the $t \times t$ determinant of a Hankel matrix for $t = \text{poly}(n)$.

Failure of Prior Strategy

In the general determinantal, skew-symmetric, and symmetric cases we analyzed the behavior of $E_{i,j}(\lambda)$.

Failure of Prior Strategy

In the general determinantal, skew-symmetric, and symmetric cases we analyzed the behavior of $E_{i,j}(\lambda)$.

This no longer works:

$$\begin{pmatrix} 1 & \lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_4 \\ x_3 & x_4 & x_5 \end{pmatrix} = \begin{pmatrix} x_1 + \lambda x_2 & x_2 + \lambda x_3 & x_3 + \lambda x_4 \\ x_2 & x_3 & x_4 \\ x_3 & x_4 & x_5 \end{pmatrix}$$

Thank You!

If you don't know how to answer someone's question, you should compliment them by saying it's a good question.

— Noga Alon

Bibliography I



Robert Andrews and Michael A. Forbes.

Ideals, determinants, and straightening: Proving and using lower bounds for polynomial ideals.

In *STOC '22: 54th Annual ACM SIGACT Symposium on Theory of Computing, Rome, Italy, June 20 - 24, 2022*, pages 389–402. ACM, 2022.



Winfried Bruns and Aldo Conca.

Gröbner bases and determinantal ideals.

In *Commutative Algebra, Singularities and Computer Algebra*, pages 9–66. Springer Netherlands, 2003.

Bibliography II



Peter Bürgisser.

Completeness and Reduction in Algebraic Complexity Theory.

Springer Berlin Heidelberg, 2000.



Peter Bürgisser.

The complexity of factors of multivariate polynomials.

Foundations of Computational Mathematics, 4(4):369–396, September 2004.

Bibliography III



Aldo Conca.

Straightening Law and Powers of Determinantal Ideals of Hankel Matrices.

Advances in Mathematics, 138(2):263–292, March 1998.



Corrado de Concini, David Eisenbud, and Claudio Procesi.

Young diagrams and determinantal varieties.

Inventiones mathematicae, 56(2):129–165, 1980.



Corrado de Concini, David Eisenbud, and Claudio Procesi.

Hodge Algebras, volume 91 of *Astérisque*.

Société mathématique de France, 1982.

Bibliography IV



Anakin Dey and Zeyu Guo.

Debordering closure results in determinantal and pfaffian ideals.

ITCS, 2025.



Pranjal Dutta and Vladimir Lysikov.

Recent Advances in Debordering Methods, 2025.



Peter Doubilet, Gian-Carlo Rota, and Joel Stein.

On the foundations of combinatorial theory: Ix combinatorial methods in invariant theory.

Studies in Applied Mathematics, 53(3):185–216, 1974.

Bibliography V



David Eisenbud.

Introduction to algebras with straightening laws.

In *Ring Theory and Algebra III*, volume 55, pages 243–268. Marcel Dekker, Inc., 1980.



Joshua A. Grochow.

Complexity in ideals of polynomials: Questions on algebraic complexity of circuits and proofs.

Bulleting of the EATCS, 131, 2020.

Bibliography VI



Adam R. Klivans and Daniel Spielman.

Randomness efficient identity testing of multivariate polynomials.

In Proceedings of the 33rd Annual ACM Symposium on Theory of Computing, pages 216–223, 2001.



Rufus Oldenburger.

Polynomials in Several Variables.

41(3):694–710, 1940.

Bibliography VII



A. Ostrowski.

On Two Problems in Abstract Algebra Connected with Horner's Rule.

In *Studies in Mathematics and Mechanics*, pages 40–48. Academic Press, 1954.



Pan.

Methods of computing values of polynomials.

21(1):105–136, 1966.